

Introduction To ROS MoveIt

Short Questions:

1. Define MoveIt and state its main purpose in ROS.
2. Explain the main functionalities of MoveIt.
3. Describe the differences between MoveIt 1 and MoveIt 2.
4. Discuss the role of motion planning in MoveIt.
5. Explain how inverse kinematics is used in MoveIt.
6. Describe the 3D perception and collision-checking capabilities of MoveIt.
7. Discuss the major research libraries used in MoveIt.
8. Compare global planners and local planners in MoveIt.
9. Explain how parallel planning architecture works in MoveIt 2.
10. Analyze how MoveIt 2 selects the best motion plan from multiple planners.
11. Compare MoveIt and RViz.
12. Compare MoveIt and ROS 2.
13. Analyze the advantages of using MoveIt with ROS 2.
14. Design a MoveIt-based robotic system for a pick-and-place task.
15. Draw and explain the basic architecture of MoveIt.

Case Based Questions:

Robot Arm Motion Planning

A manufacturing company wants to use a robotic arm to move objects between two locations on a factory floor. The robot must determine a safe path through a cluttered environment while avoiding obstacles. The engineers decide to use MoveIt because it provides motion planning, manipulation, inverse kinematics, collision checking, and trajectory execution. They also need the robot to calculate suitable joint positions for the desired end-effector pose.

Questions:

1. Explain how MoveIt can plan a safe trajectory for the robotic arm.
2. Describe how inverse kinematics can determine the required joint positions for the desired end-effector pose.
3. Analyze how collision checking helps the robot avoid obstacles during motion.

Global and Local Planning

A mobile manipulator is operating in a large environment containing several obstacles. First, the robot needs to determine a complete path from its current position to a target location. During execution, however, the robot needs to make small adjustments to maintain smooth and safe movement around nearby obstacles. The development team decides to use both global and local planning approaches.

Questions:

1. Compare the roles of global and local planners in this robotic system.
2. Describe how a global planner can determine the complete path from start to goal.
3. Explain how a local planner can make real-time adjustments during execution.

Parallel Planning in MoveIt 2

A robotics company wants its robot to find a high-quality trajectory as reliably as possible. Instead of depending on a single planning algorithm, the company uses MoveIt 2's parallel planning architecture. A single motion-plan request containing the start position, goal position, obstacles, joint limits, and other constraints is sent to several planning pipelines. OMPL, CHOMP, and PILZ independently generate candidate solutions. The system then evaluates the results and selects the most suitable trajectory.

Questions:

1. Explain how multiple planners work in parallel to solve the same motion-planning problem.
2. Describe the information contained in a MotionPlan Request and MotionPlan Response.
3. Analyze how MoveIt 2 can select the best trajectory from multiple candidate solutions.

Vision-Based Pick and Place

A warehouse robot must automatically pick objects from a container and place them at another location. The robot is equipped with a depth camera and a gripper. The system needs to identify suitable grasp poses from 3D sensor data and then plan collision-free motion for the robot. The development team uses MoveIt together with deep-learning-based grasp detection to improve the robot's ability to handle different object positions.

Questions:

1. Explain how deep learning can be used to generate grasp poses for the robot.
2. Describe how MoveIt can use the generated grasp poses for pick-and-place planning.
3. Analyze the importance of collision-free trajectory planning in this application.

Industrial Robot with ROS 2

An industrial company is developing a new robotic system that needs fast communication, reliable sensor data, multiple robot coordination, and responsive motion planning. The company chooses ROS 2 with MoveIt 2 because ROS 2 provides DDS-based communication, Quality of Service, multi-threading, modern component-based architecture, and improved support for multiple robots. The system must also respond to changing sensor information and perform safe motion planning.

Questions:

1. Discuss the advantages of using MoveIt with ROS 2 for the industrial robotic system.
2. Analyze how ROS 2 improves the performance, responsiveness, and reliability of MoveIt.
3. Explain how ROS 2 supports multi-robot communication and sensor integration in MoveIt.